```
import numpy as np
import matplotlib.pyplot as plt
```

```
# Universe of discourse
x = np.linspace(0, 10, 500)
```

```
# ── 1. Triangular Membership Function ────────────────────────
def triangular(x, a, b, c):
   left  = np.where((b - a) != 0, (x - a) / (b - a), 0)
   right = np.where((c - b) != 0, (c - x) / (c - b), 0)
   return np.maximum(np.minimum(left, right), 0)
```

```
# ── 2. Trapezoidal Membership Function ───────────────────────
def trapezoidal(x, a, b, c, d):
  left  = np.where((b - a) != 0, (x - a) / (b - a), 1)
  right = np.where((d - c) != 0, (d - x) / (d - c), 1)
  return np.maximum(np.minimum(np.minimum(left, 1), right), 0)
```

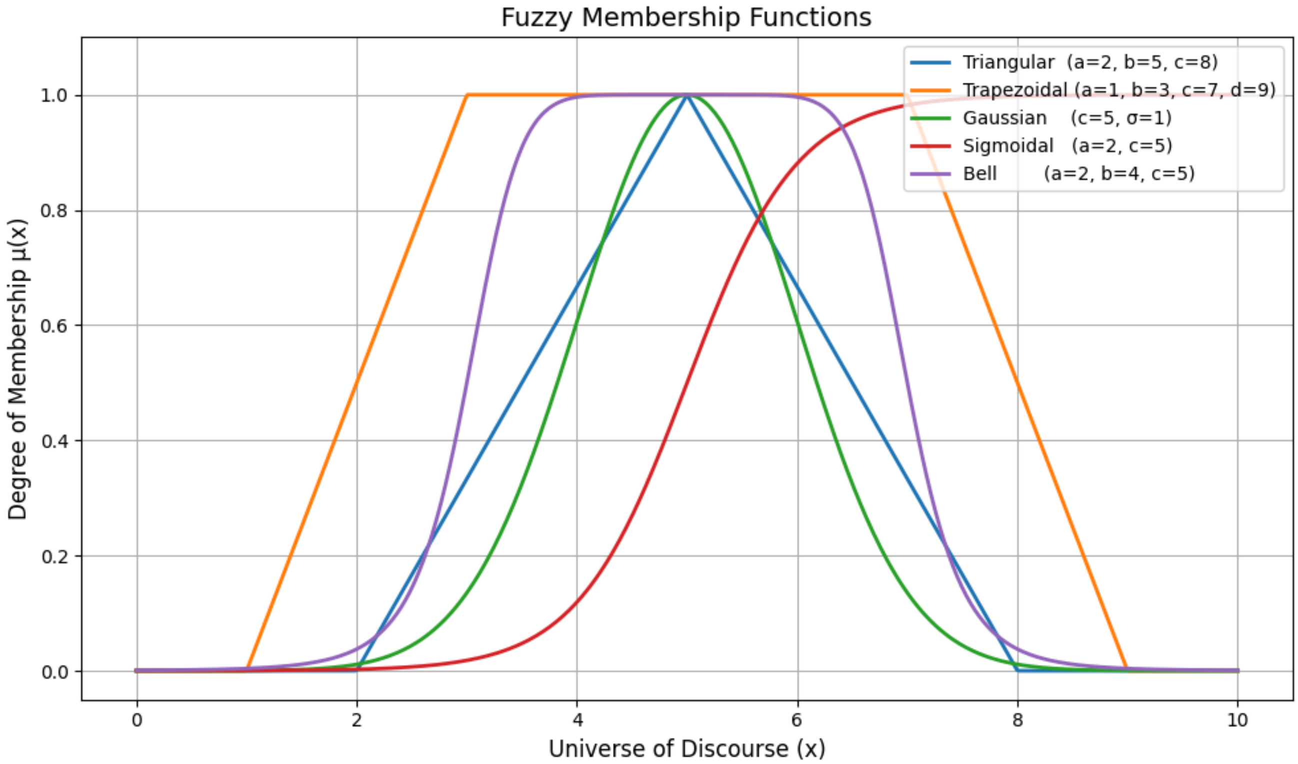
```
# ── 3. Gaussian Membership Function ──────────────────────────
def gaussian(x, c, sigma):
 return np.exp(-((x - c) ** 2) / (2 * sigma ** 2))
```

```
# ── 4. Sigmoidal Membership Function ─────────────────────────
def sigmoidal(x, a, c):
  return 1 / (1 + np.exp(-a * (x - c)))
```

```
# ── 5. Generalized Bell Membership Function ───────────────────
def bell(x, a, b, c):
  return 1 / (1 + np.abs((x - c) / a) ** (2 * b))
```

```
# ── Compute membership values ─────────────────────────────────
tri  = triangular(x, 2, 5, 8)
trap = trapezoidal(x, 1, 3, 7, 9)
gaus = gaussian(x, 5, 1)
sig  = sigmoidal(x, 2, 5)
bel  = bell(x, 2, 4, 5)
```

```
# ── Plot ──────────────────────────────────────────────────────
plt.figure(figsize=(10, 6))
plt.plot(x, tri,  label='Triangular  (a=2, b=5, c=8)',       linewidth=2)
plt.plot(x, trap, label='Trapezoidal (a=1, b=3, c=7, d=9)',  linewidth=2)
plt.plot(x, gaus, label='Gaussian    (c=5, σ=1)',            linewidth=2)
plt.plot(x, sig,  label='Sigmoidal   (a=2, c=5)',             linewidth=2)
plt.plot(x, bel,  label='Bell        (a=2, b=4, c=5)',        linewidth=2)
plt.xlabel('Universe of Discourse (x)', fontsize=12)
plt.ylabel('Degree of Membership μ(x)', fontsize=12)
plt.title('Fuzzy Membership Functions', fontsize=14)
plt.legend(loc='upper right')
plt.grid(True)
plt.ylim(-0.05, 1.1)
plt.tight_layout()
plt.savefig('membership_functions.png', dpi=150)
plt.show()
```



Start coding or generate with AI.